EMS602U-Simulation Tools in Engineering Analysis and Design

COMPUTATIONAL STRUCTURAL DYNAMICS SIMULATION WITH FINITE ELEMENT ANALYSIS

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ABSTRACT

A Finite Element Analysis (FEA) of a 2D T-shaped bracket made of mild steel was conducted using Abaqus software to assess its structural performance under a downward load of *31.23 MPa*. The stress and displacement distributions were analysed, revealing bolt holes and filleted edges as areas of high stress concentrations and structural deformation, which could lead to structural failure. A mesh convergence analysis was conducted using linear and quadratic polynomial basis functions for triangular and quadrilateral elements to study the impact of mesh refinement and element types on solution convergence and accuracy. Results demonstrated that quadratic quadrilateral elements provided the most reliable solutions for stress and displacement values, as they converged to their 'true' values of approximately *138 MPa* and *0.0342 mm*, respectively. The analysis also concluded that increasing mesh refinement and polynomial order produces more accurate and reliable FEA results. Improvement recommendations include exploring 3D modelling and validating the simulation with experimental data.

INTRODUCTION

This report presents a Finite Element Analysis of a 2D T-shaped bracket with similar dimensions to the Simpson's Strong-Tie 66T Flat T-shaped bracket [1], using the Abaqus software. The bracket is commonly used in construction to connect vertical columns to horizontal beams in timber frameworks. This analysis investigates the bracket's stress and displacement distributions when subjected to a downward load at the bottom bolt hole and assesses potential stress concentrations that may affect the bracket's structural integrity. The bracket is made of mild steel, with an average Young's Modulus, E of *210,000 MPa* [2] and a Poisson's Ratio, v of *0.303* [3]. Fixed constraints were applied to the two bolt holes on the horizontal flange of the T-bracket by applying boundary conditions in Abaqus, simulating attachment to the beam. A uniform load of *9810 N* was applied downward to the bottom bolt hole to simulate the load that the bracket supports from the horizontal beam. This is around the maximum load that T-shaped brackets of this geometry can typically support during their use in construction and corresponds to 250kg of mass [4]. This was achieved in Abaqus by applying *31.23 MPa* of surface traction to the bolt hole on the bottom of the vertical stem of the bracket. The bracket was modelled as a 2D plane stress problem with a focus on the stress and displacement distributions across its structure. The geometry of the bracket includes the following dimensions:

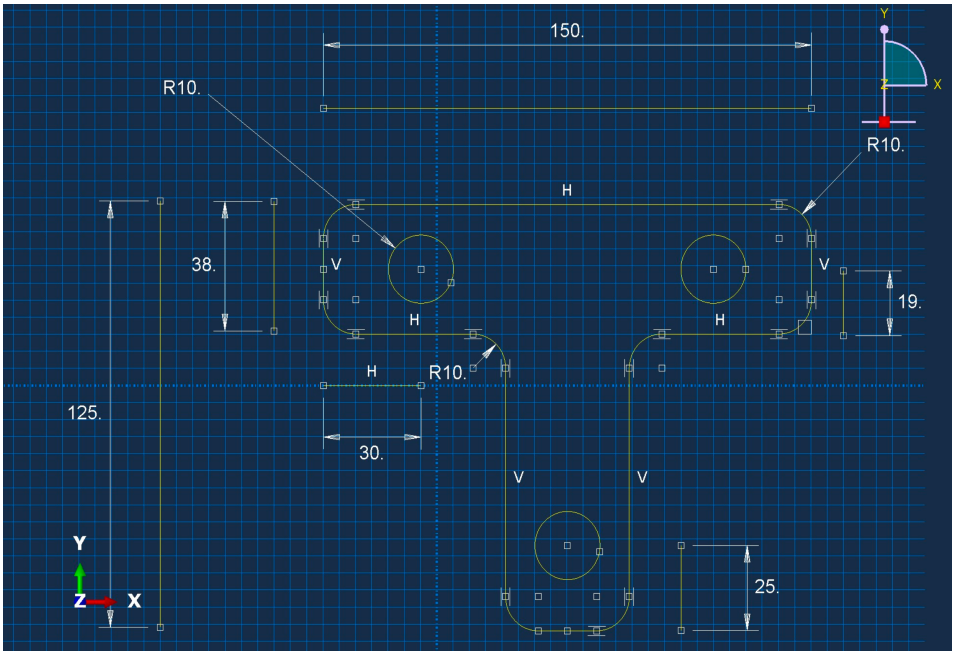


Figure 1- Abaqus sketch of T shaped bracket

Dimension	Magnitude (mm)
Total Length	*150*
Height	*125*
Width	*38*
Thickness	*2*
Radii of Bolt Holes	*10*
Fillet Radii	*10*

Table 1- Dimensions of T-shaped bracket

RESULTS

Element size	Number of elements	Stress, Mises (MPa)	Displacement magnitude(mm)
8	284	69.50	0.2901
4	1,176	95.03	0.03254
2	4,181	113.7	0.03365
1	16,745	130.1	0.03407
0.5	66,316	130.6	0.03418
0.4	103,003	131.9	0.03419

Table 2- linear triangular elements

Element size	Number of elements	Stress, Mises (MPa)	Displacement magnitude(mm)
8	168	71.44	0.03567
4	664	92.76	0.03394
2	2,499	108.9	0.03471
1	9,743	118.5	0.03420
0.5	39,469	129.0	0.03422
0.4	N/A	N/A	N/A

Table 3- linear quadrilateral elements

Element size	Number of elements	Stress, Mises (MPa)	Displacement magnitude(mm)
8	284	116.6	0.03381
4	1,176	132.6	0.03416
2	4,181	136.7	0.03421
1	16,745	137.0	0.03422
0.5	66,316	138.2	0.03422
0.4	103,003	138.1	0.03422

Table 4- quadratic triangular elements

Element size	Number of elements	Stress, Mises (MPa)	Displacement magnitude(mm)
8	168	144.2	0.03399
4	664	141.4	0.03420
2	2,499	137.1	0.03422
1	9,743	137.6	0.03422
0.5	39,469	138.0	0.03422
0.4	N/A	N/A	N/A

Table 5- quadratic quadrilateral elements

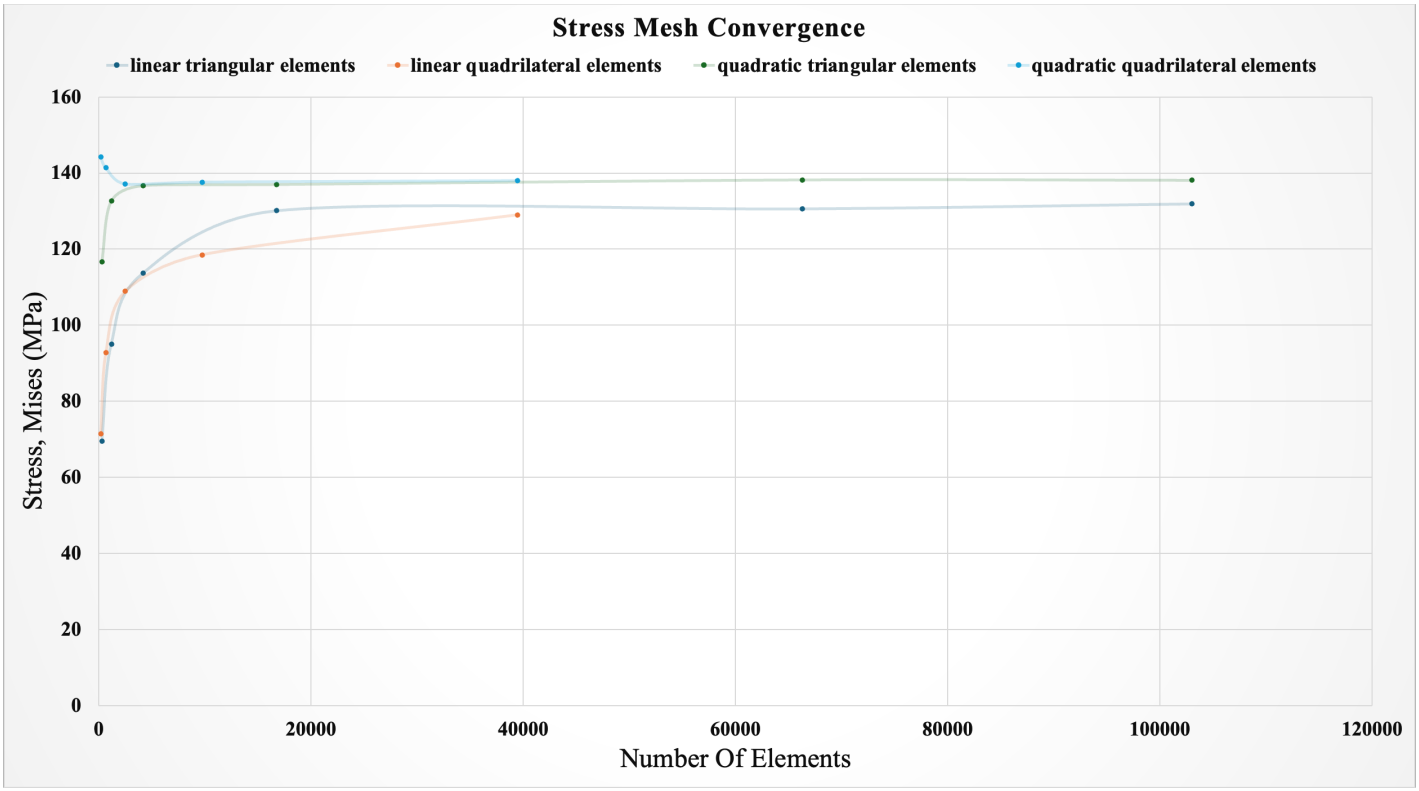


Figure 2- Stress Convergence of all mesh configurations

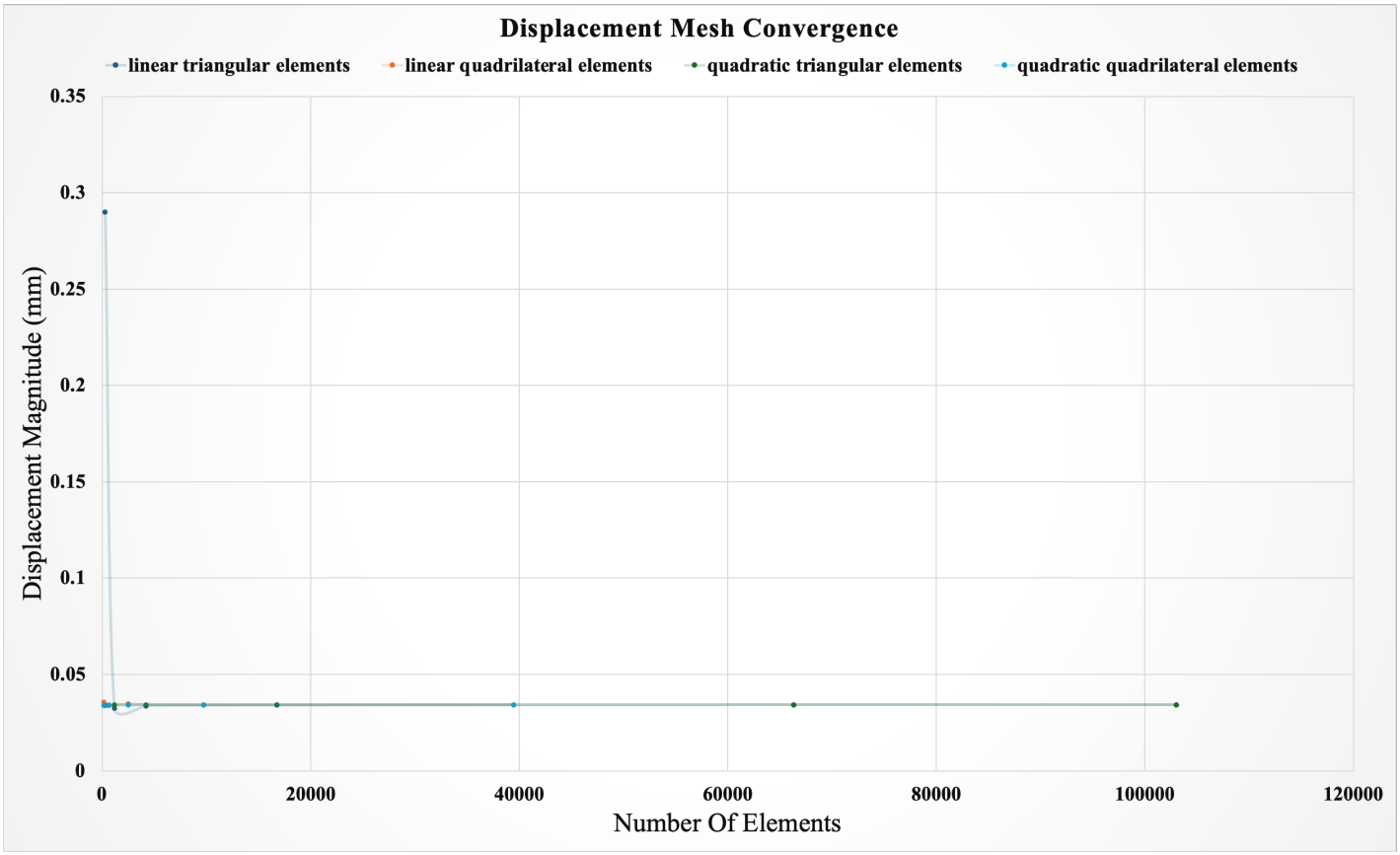


Figure 3- Displacement Convergence of all mesh configurations

Mesh Convergence Analysis

The mesh convergence analysis was carried out after applying the *31.23 MPa* of surface traction to the bottom bolt hole of the bracket and visualising the consequent deformation on Abaqus. In the visualisation module, Abaqus auto-computed a deformation scale factor of 440.2 to amplify and enhance the clarity of the deformation, and a contour plot was displayed which revealed the areas experiencing the highest stress and displacement concentrations due to the load. *Figure 4* shows that the highest stress concentrations were found around some edges of the bolt holes, on the bottom bolt hole where the load was directly applied, and on the filleted edges around the transition from the horizontal to the vertical sections of the T bracket, with the maximum stress occurring at the edge of the bottom bolt hole. These stress concentrations typically occur in structural components where geometrical irregularities, such as holes, cause an increase in working stress compared to surrounding areas. These local stress areas are likely to contribute to the quick failure of structures [5]. *Figure 5* also shows that the highest displacement magnitudes occur at the regions near the bottom bolt hole area, and a gradual decrease in displacement is seen climbing up towards the centre of the T stem. This shows that these areas experience the highest deformations due to stress and are regions where structural failures in the T-bracket may occur.

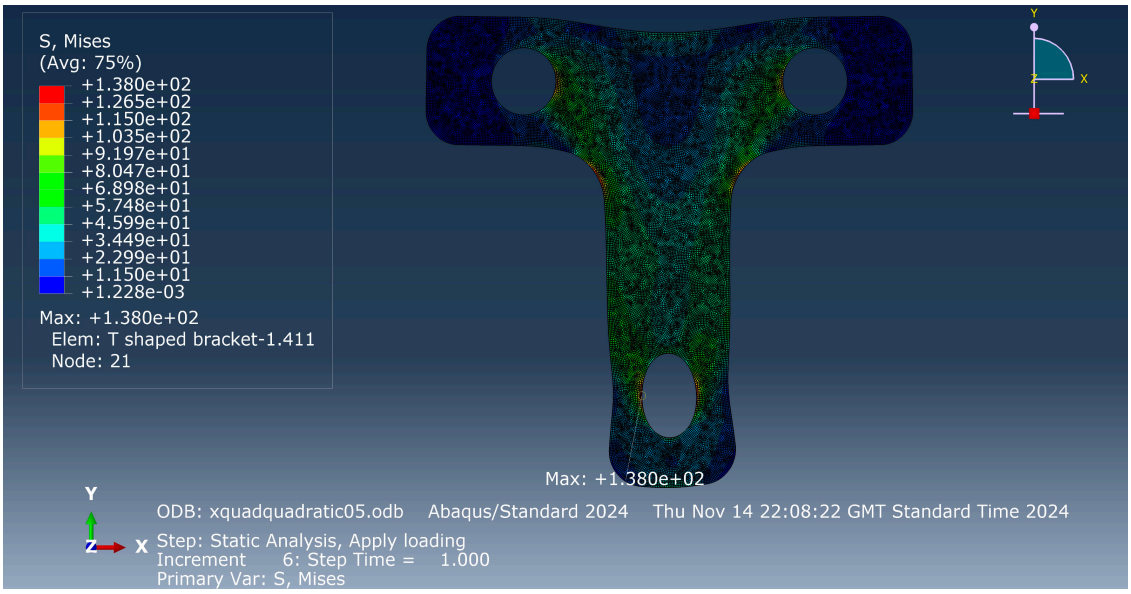


Figure 4- Contour plot of Von Mises Stress (MPa) on the T shaped bracket

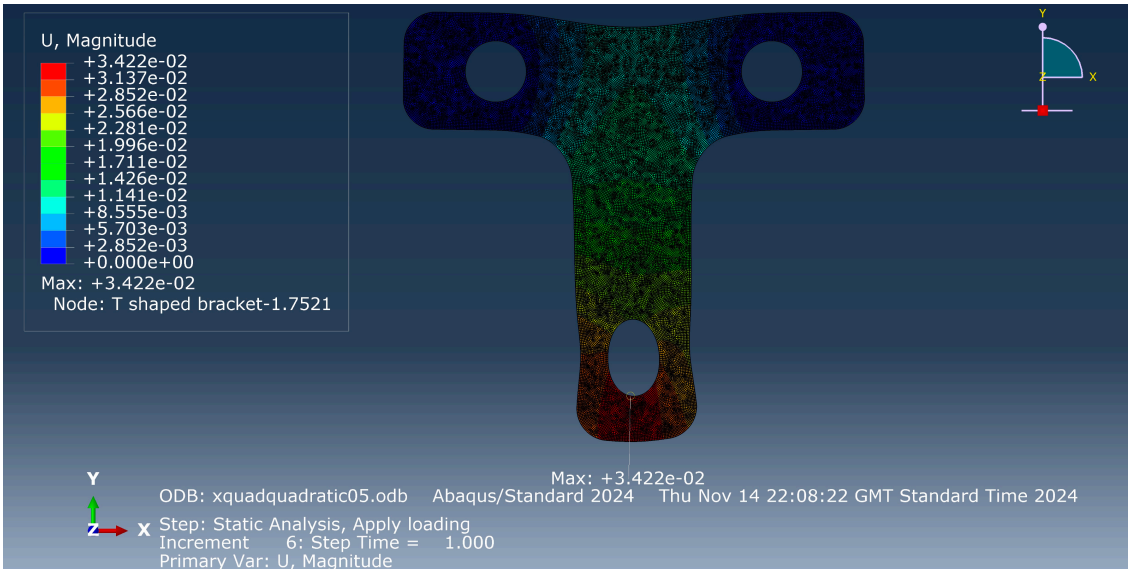


Figure 5 – Contour plot of Displacement Magnitudes (mm) in the T-shaped bracket

The Von Mises Stress and Displacement convergence analyses were carried out by first calculating an initial mesh size for the T-bracket based on the *10 mm* radius of the bolt holes, which are the primary areas of interest where structural problems are expected to occur. The curve length of *62.83 mm* was divided by 4, then by 2, resulting in an initial mesh size of approximately *8 mm*. The initial coarse meshes were then progressively refined by halving the element size each step. For most mesh configurations, the finest mesh size could be computed as 0.4, but due to license limitations, some mesh element sizes were only computed down to 0.5. This analysis was carried out using two main meshing element types, triangular and quadrilateral shaped elements, as well as two polynomial order basis functions, linear and quadratic, resulting in four different mesh configurations as follows: CPS8R: quadratic quadrilateral (8 nodes), CPS6M: quadratic triangular (6 nodes), CPS3: linear triangular (3 nodes), and CPS4R: linear quadrilateral (4 nodes). The number of elements produced, the maximum Von Mises stress, and the maximum displacement values corresponding to each mesh configuration were then computed in Abaqus with decreasing element sizes for each configuration and recorded in *Tables 2-5*.

Effect of Mesh Refinement

In all mesh configurations, the stress and displacement seemed to approach stable values with increasingly finer meshes. For each mesh configuration, the stress values started at lower values for coarser meshes, increased progressively with increasing mesh size and then began to stabilise as the stress converged towards its 'true value' which is *138 MPa* based on the results from this analysis. This is shown in the stress convergence graph in *Figure 2*. Displacement values also show a similar trend, converging to its 'true value' of *0.0342 mm* as the mesh size increased in all mesh configurations. This is shown in *Figure 3*. The more refined meshes showed results that were closer to the 'true value', indicating that a sufficiently fine mesh is required to accurately capture displacement and stress distribution and to achieve reliable results in Finite Element Analysis.

Comparing polynomial orders of basis functions

The solution convergence for the triangular elements as displayed in *Table 2* shows the stress values converging from *69.5 MPa* to *131.9 MPa*, and displacement converging from *0.29010 mm* to *0.03419 mm* as the mesh refines using the linear basis function. However, for the same triangular elements using instead the quadratic basis function, the stress values started at a higher value of *116.6MPa* and converged to *138.1 MPa*, with displacement consistently converging from *0.0338 mm* to *0.0342 mm* as shown in *Table 4*. This demonstrates that the quadratic basis function caused the solution convergence of both stress and displacement to stabilise closer to their 'true value' as compared to the linear basis function. Based on these results, it can be concluded that the quadratic basis function produces better convergence as compared to the linear. This is further proven by the results in *Tables 3 and 5* representing linear quadrilateral elements and quadratic quadrilateral elements, as both the stress and displacement values show a higher degree of convergence in the quadratic basis function for analysis using the same element.

Comparing meshing elements

The solutions for triangular and quadrilateral elements also exhibited differences in stress and displacement convergence. The stress solution convergence for the quadratic triangular elements shown in *Table 4* shows stress values converging from *116.6 MPa* to *138.1 MPa*. Initial stress however begins at a higher value of *144.2 MPa* for quadratic quadrilateral elements and then converges to *138 MPa*. Although both element types stabilise around 138 MPa, the quadratic quadrilateral elements achieve this stability more rapidly as shown in *Table 5*, with fewer fluctuations. The displacement solution convergence of the two element types also shows similar behaviour, as they both stabilise to *0.03422 mm*. However, the quadrilateral elements achieve this

stability slightly quicker and more consistently than the triangular elements. This infers that the quadrilateral elements provide a more reliable solution convergence for stress and displacement as compared to triangular elements, with less variation across mesh sizes. This idea is reinforced by the convergence behaviour of the linear triangular versus linear quadrilateral mesh configurations, with the quadrilateral elements generally showing better and more stable stress and displacement convergence across mesh sizes.

Based on these results, quadratic quadrilateral elements (8 nodes) offer the most reliable solution convergence in both stress and displacement values, followed by quadratic triangular elements (6 nodes), linear quadrilateral elements (4 nodes), and then triangular linear elements (3 nodes), which show the least stable stress and displacement solution convergence. This ranking suggests that the more nodes used in Finite Element Analysis, the more stable the solution convergence, and the greater the reliability of the results.

CONCLUSION

This report entails a Finite Element Analysis of a 2D T-shaped bracket made of mild steel under a downward load of *31.23 MPa* applied at the bottom bolt hole using Abaqus software. The analysis revealed that the greatest stress concentrations occur around the bolt holes and filleted edges of the bracket, with the maximum stress occurring at the bottom bolt hole where the load was directly applied. The maximum displacement was also observed around the bottom bolt hole, with a gradual upward decrease towards the centre of the bracket, showing the structural deformation in the areas of high stress. These areas are thus proven to be the primary areas where structural failures are most likely to occur due to abrupt geometry changes which led to localised stress concentrations. Mesh refinement greatly impacted the accuracy of the results. As the mesh was refined, stress and displacement values converged toward stable ‘true’ values of approximately *138 MPa and 0.0342 mm* respectively. This demonstrates that achieving accurate results in FEA requires a sufficiently fine mesh and that the results achieved in this analysis were reliable. Between the linear and quadratic polynomial basis functions, quadratic configurations showed faster convergence at higher, more stable stress values compared to linear configurations, thus providing more reliable solution convergence than linear basis functions. Between the two meshing element types, quadrilateral elements displayed better solution convergence than triangular elements. Overall, the quadratic quadrilateral mesh configuration achieved the most consistent and reliable solution convergence in both stress and displacement values among all the other mesh configurations. The overall observation suggested that increasing the number of nodes used in Finite Element Analysis causes better solution convergence and more reliable results.

Future recommendations to further improve the accuracy and reliability of this analysis include Extending the study to analyse a 3D model of the bracket and exploring more meshing element types for a more realistic understanding of the bracket’s behaviour. Focusing the mesh refinement on the areas of interest such as the fillets and bolt holes to minimise the computational cost of the analysis and comparing the simulation results with physical testing results to ensure the simulation reflects real-world performance could also be good ways to improve the accuracy and reliability of this analysis.

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